SCHOOL OF APPLIED SCIENCE AND OPEN LEARNING

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF CYBER SECURITY

OPERATING SYSTEM SECURITY
CYS 337

SESSIONAL EXAMINATION

DATE: 07-Nov-24 TIME: 09.00 - 12.00 HRS, 3 HOURS

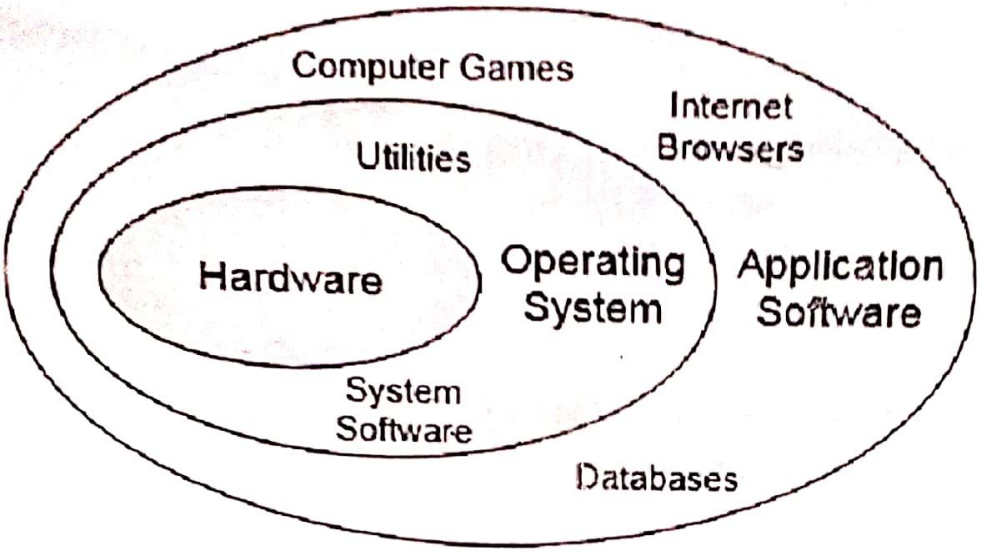
Instructions to candidates

1. Write your **Name** and **Student ID** on the **Answer Sheet** provided
2. There are **Six (6) questions** in this paper. Attempt only **five (5).** Indicate on the Answer sheet your choice from the optional questions.
3. Write your Answers on the answer Sheet provided.
4. No Foreign material will be allowed in this examination.
5. The number of marks is shown in brackets **[ ]** at the end of each question or part question.
6. Clear handwriting and neat work is encouraged.
7. The total number of marks for this paper is **100**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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QUESTION 1



The diagram above shows an operating system layer from the hardware

a) Explain each of the following types of software below

i)	OS Utilities	2 Marks
ii)	Operating System	2 Marks
iii)	System software	2 Marks
iv)	Databases	2 Marks
v)	Internet browsers	2 Marks
vi)	Computer Games	2 Marks

b) What suitable general mitigations can you apply to SECURE above software — 8 Marks

Total 20 Marks

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QUESTION 2

Operating system security involves maintaining the operating system

a) Explain the role of the following OS security maintenance requirements:
   i) Patches — 5 Marks
   ii) Fixes — 3 Marks
   iii) Revisions — 2 Marks

b) Explain the roles of the following in maintaining OS security
   i) Security Maintenance Strategy — 5 Marks
   ii) Post-Install Security Checklist — 5 Marks

Total 20 Marks

QUESTION 3

a) Explain what a File System is, giving examples — 4 Marks

c) Describe how the OS file system help secure the disk, folders and files through permissions, access and file share security — 12 Marks

d) Explain Files System Audits and monitoring — 4 Marks

Total 20 Marks

QUESTION 4

One of the ways to allow and manage users access a particular network securely is through implementation of Active Directory and Domain Services (AD DS) - for windows domain controllers

a) What is the AD, DS — *2 Marks*
b) What is a domain controller (DC) why are they important — *4 Marks*
c) Explain the terms
   i) DNS — *2 Marks*
   ii) Forest — *2 Marks*
   iii) Replication — *2 Marks*

d) Describe how the AD co-operates with DNS during the User logon process — *8 Marks*

Total 20 Marks

QUESTION 5

During logon process the Operating system may employ various ways of authentication depending on the sensitivity of the platform being logged on to:

Explain the following Operating System Authentication types

i) Password Based Authentication — 4 Marks
ii) Multi-factor Authentication — 4 Marks
iii) Certificate – based Authentication — 4 Marks
iv) Token Based Authentication — 4 Marks
v) Biometric based Authentication — 4 Marks

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Total 20 Marks

QUESTION 6

Cloud security is a collection of procedures, responsibilities and technology to secure the external and internal threats to cloud infrastructure.

a) Outline TWO shared responsibilities between the Cloud Provider and the Cloud Client in ensuring the protection of the service provided and the cloud infrastructure as a whole. 8 Marks

b) Give FOUR examples of Cloud providers around the World and one example of such in Zambia 4 Marks

c) State FOUR benefits of migrating IT-infrastructure to cloud environment as opposed to having an "on-premise" IT-Infrastructure. 4 Marks

d) In terms of Cloud Service Models explain SecaaS as a cloud service

4 Marks

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Total 20 Marks

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